

RESEARCH INTERESTS

My current research centers on **Embodied AI** and **Continual Learning**, particularly latent world models, test-time training, and persistent memory.

Areas: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Cognitive Science, Developmental Psychology

EDUCATION

Johns Hopkins University, Baltimore, MD, US

08/2025 - 05/2029,2030 (expected)

Ph.D. in Computer Science

Advisor: Tianmin Shu, Daniel Khashabi

University of Michigan, Ann Arbor, MI, US

08/2022 - 04/2024

M.S. in Robotics

Certificate, Cognitive Science

GPA: 4.00/4.00

Advisor: Joyce Chai

University of Massachusetts Amherst, Amherst, MA, US

09/2018 - 05/2022

B.S. in Computer Science

GPA: 3.85/4.00

PUBLICATIONS

* → equal contribution. † → equal advising.

- [P.1] Yifan Yin*, Zehao Wen*, Suyu Ye, Jieneng Chen, Zehan Zheng, Nanru Dai, Haojun Shi, Aydan Huang, **Zheyuan Zhang**, Alan Yuille, Jianwen Xie, Ayush Tewari, Tianmin Shu. “3D-Belief: Embodied Belief Inference via Generative 3D World Modeling”. In *arXiv*. 2026.
- [W.1] **Zheyuan Zhang**, Suyu Ye, Nakul Agarwal, Hossein Nourkhiz Mahjoub, Ehsan Moradi Pari, Daniel Khashabi, Tianmin Shu, Vaishnav Tadiparthi. “JEPA-TTT: Persistent Test-Time Training of Latent World Models for Planning under Dynamics Shifts”. In *WM PAI Workshop @ NeurIPS*. 2026.
- [W.2] Suyu Ye, **Zheyuan Zhang**, Vaishnav Tadiparthi, Hossein Nourkhiz Mahjoub, Ehsan Moradi Pari, Tianmin Shu, Homanga Bharadhwaj, Nakul Agarwal. “Watch, Infer, Coordinate: Inferring Robot Partner Constraints for Zero-Shot Coordination”. In *PhysUnderstand Workshop @ NeurIPS*. 2026.
- [W.3] **Zheyuan Zhang***, Alvin Zhang*, Daniel Khashabi†, Tianmin Shu†. “Continual Learning Mechanisms Compose for Long-Horizon Memorization”. In *CL4FMAgents Workshop @ NeurIPS*. 2026.
- [W.4] Hongxin Zhang*, **Zheyuan Zhang***, Zeyuan Wang*, Zunzhe Zhang, Lixing Fang, Qinhong Zhou, Chuang Gan. “Ella: Embodied Lifelong Learning Agents with Non-Parametric Memory”. In *arXiv*. 2026.
- [C.1] **Zheyuan Zhang***, Zehao Wen*, Alvin Zhang, Andrew Wang, Jianwen Xie, Daniel Khashabi†, Tianmin Shu†. “AgentOdyssey: Open-Ended Long-Horizon Text Game Generation for Test-Time Continual Learning Agents”. In *Advances in Neural Information Processing Systems 40 (NeurIPS)*. 2026.

- [C.2] Qinhong Zhou*, Hongxin Zhang*, Xiangye Lin*, **Zheyuan Zhang***, Yutian Chen, Wenjun Liu, Zunzhe Zhang, Sunli Chen, Lixing Fang, Qiushi Lyu, Xinyu Sun, Jincheng Yang, Zeyuan Wang, Bao Chi Dang, Zhehuan Chen, Daksha Ladia, Jiageng Liu, Chuang Gan. “Virtual Community: An Open World for Humans, Robots, and Society”. In *The Fourteenth International Conference on Learning Representations (ICLR)*. 2026.
- [C.3] **Zheyuan Zhang***, Fengyuan Hu*, Jayjun Lee*, Freda Shi, Parisa Kordjamshidi, Joyce Chai, Ziqiao Ma. “Do Vision-Language Models Represent Space and How? Evaluating Spatial Frame of Reference Under Ambiguities”. In *The Thirteenth International Conference on Learning Representations (ICLR)*, **Oral Presentation (1.8%)**. 2025.
- [C.4] Yuncong Yang*, Jiageng Liu*, **Zheyuan Zhang**, Siyuan Zhou, Reuben Tan, Jianwei Yang, Yilun Du, Chuang Gan. “MindJourney: Test-Time Scaling with World Models for Spatial Reasoning”. In *Advances in Neural Information Processing Systems 39 (NeurIPS)*. 2025.
- [C.5] Shane Storks, Itamar Bar-Yossef, Yayuan Li, **Zheyuan Zhang**, Jason J. Corso, Joyce Chai. “Transparent and Coherent Procedural Mistake Detection”. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2025.
- [C.6] Hongxin Zhang*, Zeyuan Wang*, Qiushi Lyu*, **Zheyuan Zhang**, Sunli Chen, Tianmin Shu, Behzad Dariush, Kwonjoon Lee, Yilun Du, Chuang Gan. “Compositional World Models for Embodied Multi-Agent Cooperation”. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025.
- [C.7] **Zheyuan Zhang**. “A Combinatorial Approach to Neural Emergent Communication”. In *Proceedings of the 31th International Conference on Computational Linguistics (COLING)*. 2025.
- [C.8] Keunwoo Peter Yu, **Zheyuan Zhang**, Fengyuan Hu, Shane Storks, Joyce Chai. “Eliciting In-Context Learning in Vision-Language Models for Videos Through Curated Data Distributional Properties”. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2024.
- [C.9] **Zheyuan Zhang***, Shane Storks*, Fengyuan Hu, Sungryull Sohn, Moontae Lee, Honglak Lee, Joyce Chai. “From Heuristic to Analytic: Cognitively Motivated Strategies for Coherent Physical Commonsense Reasoning”. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), Long Papers*. 2023.
- [C.10] **Zheyuan Zhang**, Huiliang Shang. “Low-cost Solution for Vision-based Robotic Grasping”. In *Proceedings of the 2021 International Conference on Networking Systems of AI (INSAI)*, **Second Prize Excellent Paper Award**. 2021.

RESEARCH EXPERIENCE

Honda Research Institute , Research Intern Adviser: Hossein Nourkhiz Mahjoub, Vaishnav Tadiparthi	San Jose, CA, US 05/2026 - 08/2026
MIT-IBM Watson AI Lab , Visiting Researcher Microsoft Research , Research Collaboration Adviser: Chuang Gan	Cambridge, MA, US 06/2024 - 05/2025
University of Michigan SLED Research Lab , Research Assistant Adviser: Joyce Chai	Ann Arbor, MI, US 11/2022 - 05/2024
LG AI Research , Research Collaboration	Ann Arbor, MI, US

Adviser: Honglak Lee

11/2022 - 11/2023

Fudan University RAS Lab, Research Intern

Shanghai, CN

Adviser: Huiliang Shang, Ruijiao Li

06/2021 - 08/2021

HONORS AND AWARDS

Gold Reviewer. International Conference on Machine Learning (ICML).	2026
Oral Presentation (1.8%). International Conference on Learning Representations (ICLR).	2025
Chancellor's Award, \$40,000 USD (total). UMass Amherst.	2018-2022
Dean's List Honors (all semesters). UMass Amherst.	2018-2022
Second Prize Excellent Paper Award. INSAI 2021 Conference.	2021

TEACHING

Graduate Student Instructor (GSI)	Winter 2024, Fall 2023
EECS 492: Introduction to Artificial Intelligence	
Topics: Search, Constraint Satisfaction Problems, Logic and Inference, Uncertainty, Bayesian Networks, Decision Trees, Linear Regression, Neural Networks and Generative AI, Nonparametric Methods, Decision Theory, Reinforcement Learning, Game Theory	
Guest Lectures	
Apr. 2024	University of Michigan, Host: Alexander Rodríguez, Laura Wendlandt
Nov. 2023	University of Michigan, Host: Emily Mower Provost, Alexander Rodríguez

ACADEMIC SERVICE

Workshop Organizing

- Leading Organizer, Towards Test-Time Continual Learning Agents (TTCL @ NeurIPS 2026).

Journal Reviewing

Machine Learning TMLR

Conference Reviewing

Machine Learning ICLR, NeurIPS, ICML
Natural Language Processing ACL

Workshop Reviewing

Natural Language Processing ACL-SRW, EACL-SRW
Machine Learning NeurIPS Pluralistic Alignment
Robotics RSS Continual Robot Learning from Humans

OTHER PROJECTS

1. Bot Lab: Autonomous Ground Vehicle from Low-level Control, SLAM to Planning and Exploration	2022
2. Clara in Wonderland: 3D Open-world Adventure Game (Unity, C#)	2021
3. Quanin: Automatic Stock Screener (Python, C#)	2021

- | | |
|---|------|
| 4. Blockchain From Scratch (C++) | 2020 |
| 5. Findurcourse.com: Node.js Web Application (HTML/CSS, JavaScript, PostgreSQL) | 2020 |
| 6. ZiZoyaOS: 32-Bit Operating System From Scratch (Assembly, C) | 2020 |

SKILLS

- Programming Languages: Python, C/C++, C#, JavaScript, MATLAB, Visual Basic, Pascal
- Other Computer Languages and Software: HTML/CSS, SQL, L^AT_EX, Blender, Unity, Unreal Engine